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Magik in action

Hence sonic vibrations allow formation of a homogenous mass devoid of any air space. Vibratory strokes enable Gutta percha to flow into lateral canals, achieving better lateral canal seal.

The cost factor

Jain discloses that cost was always a concern as his parents funded the project. He adds, "I could not think of being lavish and tried to keep the costs as low as possible without compromising on efficiency. Hence I went ahead with the use of sonic vibration, which by itself makes the product economical."

Jain created about seven prototypes before he came up with the final version. Creating the prototypes cost between Rs 2000 and Rs 3000. It was still much better than similar machines which cost Rs 100,000-125,000. Jain decided to use sonic vibrations to help Gutta percha flow in lateral canals, i.e., the horizontal plane.

"Sonic vibration is a cheaper alternative to ultrasonic. Moreover, recent studies have shown it to be more efficient," Jain explains.

The cost of creating the final prototype was about Rs 1000, but this is not the final estimate for a fully manufactured product.

The prototype has been used for over a hundred extracted tooth samples without any hardware maintenance.

Jain says that it is basically a question of proper maintenance by the operator for ensuring its optimal lifespan.

Jain has won many awards for his device at national conferences like MIT's TR35, AVISHKAR inter-university state-level research festival, Indian Dental Conference (IDC) and Scientific Rendezvous. He also represented India at Dentsply Student Clinician Programme of America Dental Association (ADA) in Florida, USA.

What lies ahead

Jain is improvising on his device to maximise its productivity and make it more compact and convenient for clinical use. He reveals, "I am working on adding injectible Gutta percha carrier system to the device for clinical use. I am also trying to replace the thermal property achieved by resistance-based system with a faster induction-based system. I am looking for proper funds and resources for the same."

Jain is conducting lab and clinical studies on the prototype and looking for companies that will sell his device.

"Since the production cost of Magik is very low, its market price will be much lower than already available products. This will result in a lower cost for root-canal treatment too," Jain concludes. ●

The author is from EFY Bureau, Bengaluru